# SuperPoint: Self-Supervised Interest Point Detection and Description

Daniel DeTone, Tomasz Malisiewicz, Andrew Rabinovich

**Research @ Magic Leap**

*CVPR 2018 Deep Learning for Visual SLAM Workshop*

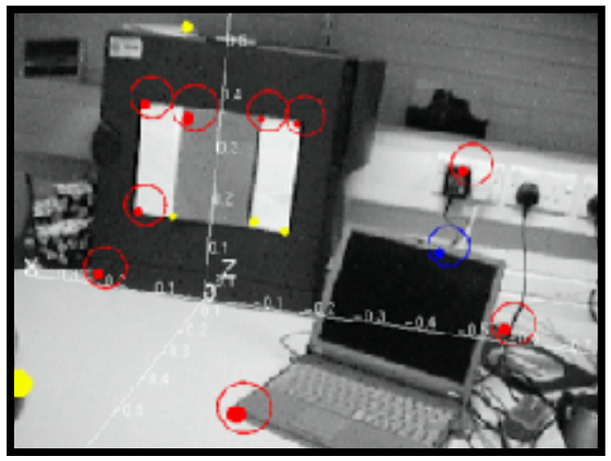
# Main Ideas

- “SuperPoint”
  - A Deep SLAM Frontend
  - Multi-task fully convolutional network
  - Designed for Real-time
- “Homographic Adaptation”
  - Self-supervised recipe to train keypoints
  - Synthetic pre-training
  - Homography-inspired domain adaptation

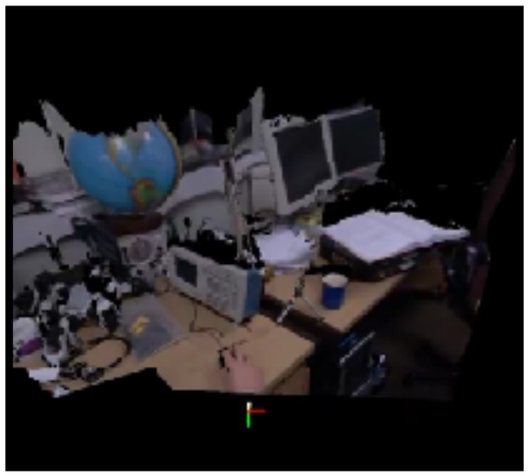
# 2000-2015 Visual SLAM

- Great Visual SLAM Research
- Real-time systems emerge
- Very few learned components

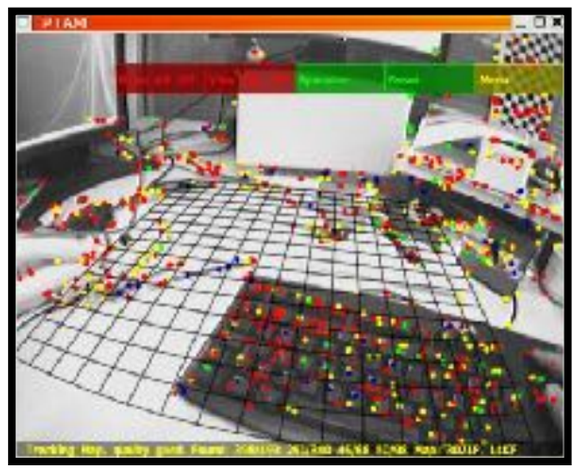
**MonoSLAM**



**DTAM**



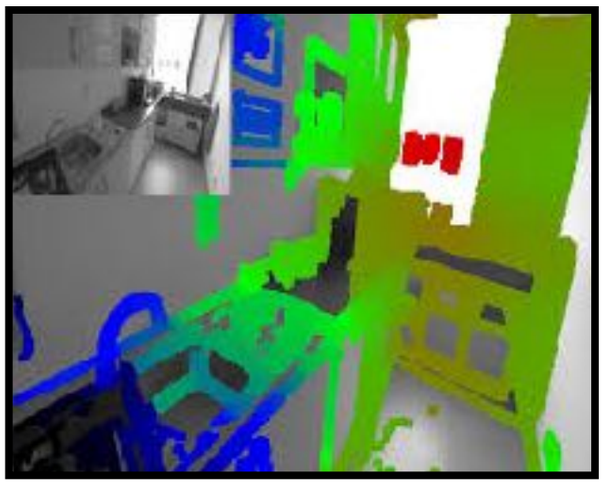
**PTAM**



**ElasticFusion**



**LSD-SLAM**



**KinectFusion**



**Event-camera SLAM**

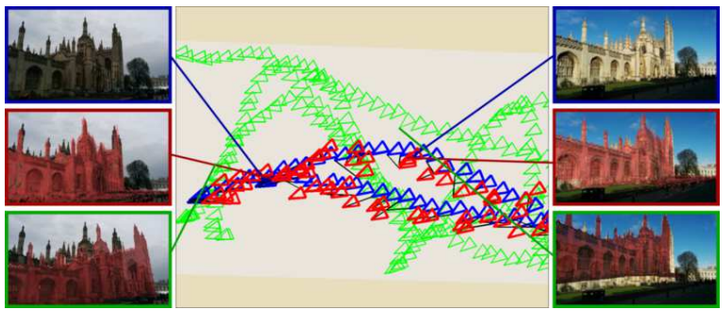


**DynamicFusion**

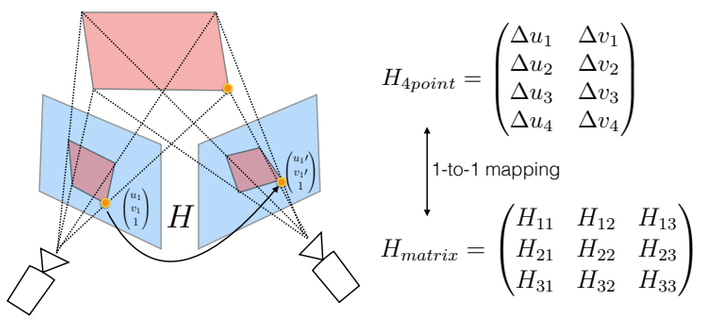


**Collage courtesy: Andrew Davison's ICCV 2015 Future of Real-time SLAM workshop talk**

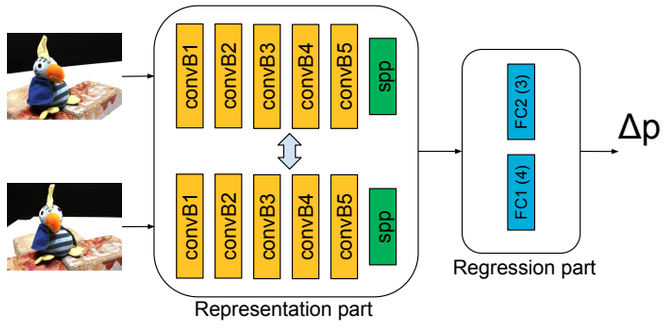
# 2015-2016: Simple End-to-End Deep SLAM?



PoseNet: A Convolutional Network for Real-time 6 DOF Localization




Deep Image Homography Estimation

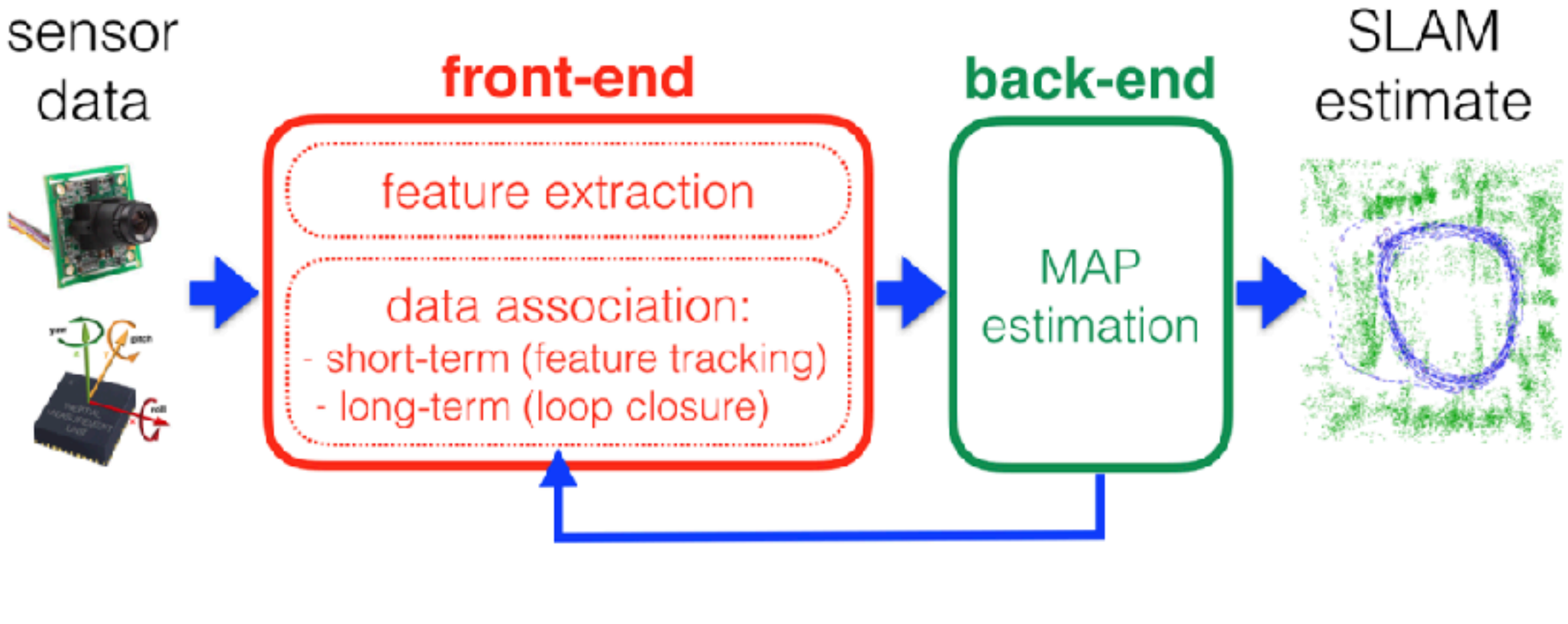



Relative Camera Pose Estimation Using Convolutional Neural Networks




- Deep Learning excitement is very high
- Simple end-to-end setups work across many computer vision tasks
  - Purely data-driven, powerful
  - Very few heuristics / little hand-tuning
- Accuracy not yet competitive
  - Maybe due to lack of large-scale data

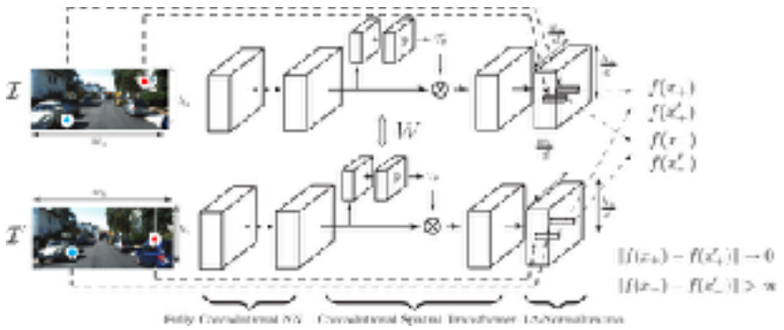
# 2017-2018: Splitting Up the Problem



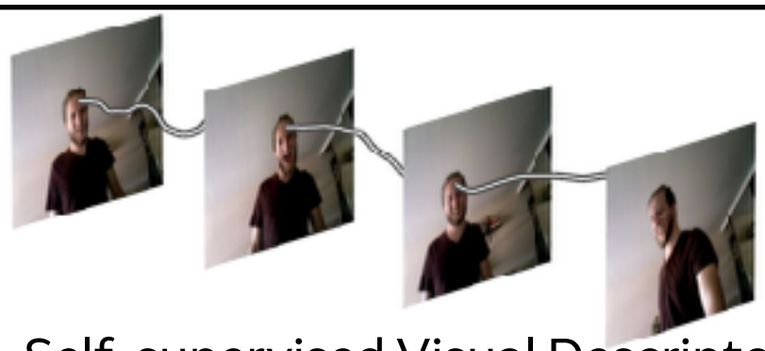

- **Frontend**: Image inputs
  - Deep Learning success: Images + ConvNets
  - Most of current work “deep-ifys” the Frontend -> Focus of this talk
- **Backend**: Optimization over pose and map quantities
  - 2018: Early deep learning work -> Focus of other oral at 12:05pm

**Photo Credit: Cadena et al 2016**

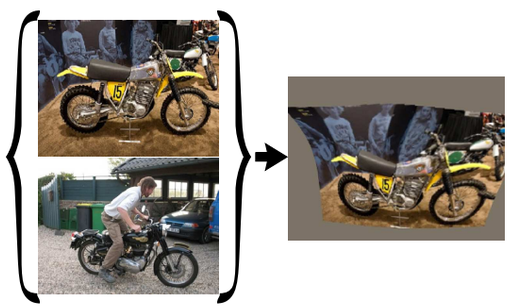
# 2017-2018 Deep Frontends: Dense



Universal Correspondence Network



Self-supervised Visual Descriptor Learning for Dense Correspondence

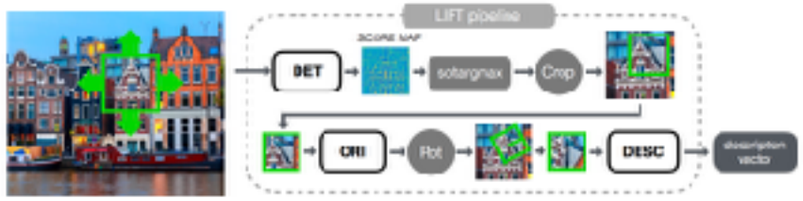


Convolutional neural network architecture for geometric matching

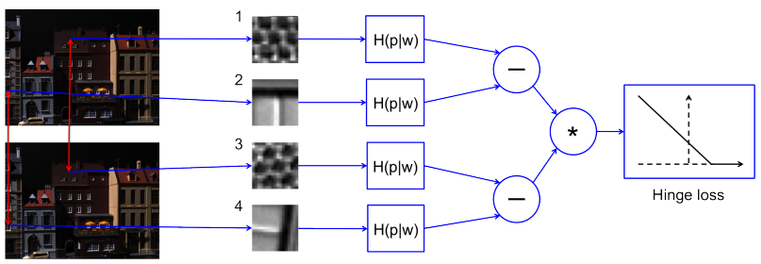



- Dense output approaches
- <u>Powerful</u> Matchability
- Not practical in low-compute SLAM systems
  - Too expensive for realtime BA

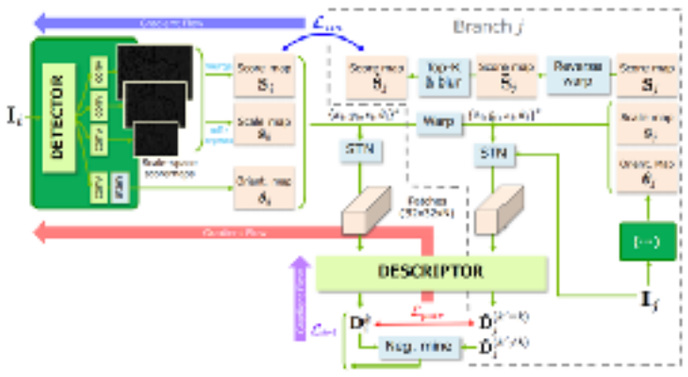
# 2017-2018 Deep Frontends: Sparse




LIFT: Learned Invariant Feature Transform

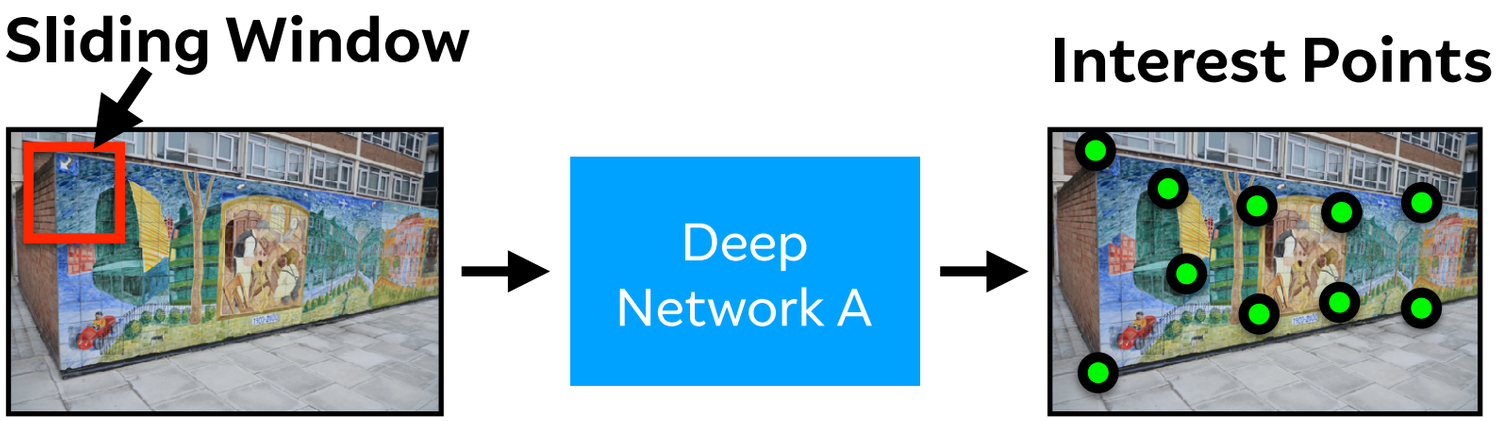



QuadNetworks: Unsupervised Learning to Rank for Interest Point Detection




LF-Net: Learning Local Features from Images

## Existing Patch-based Systems

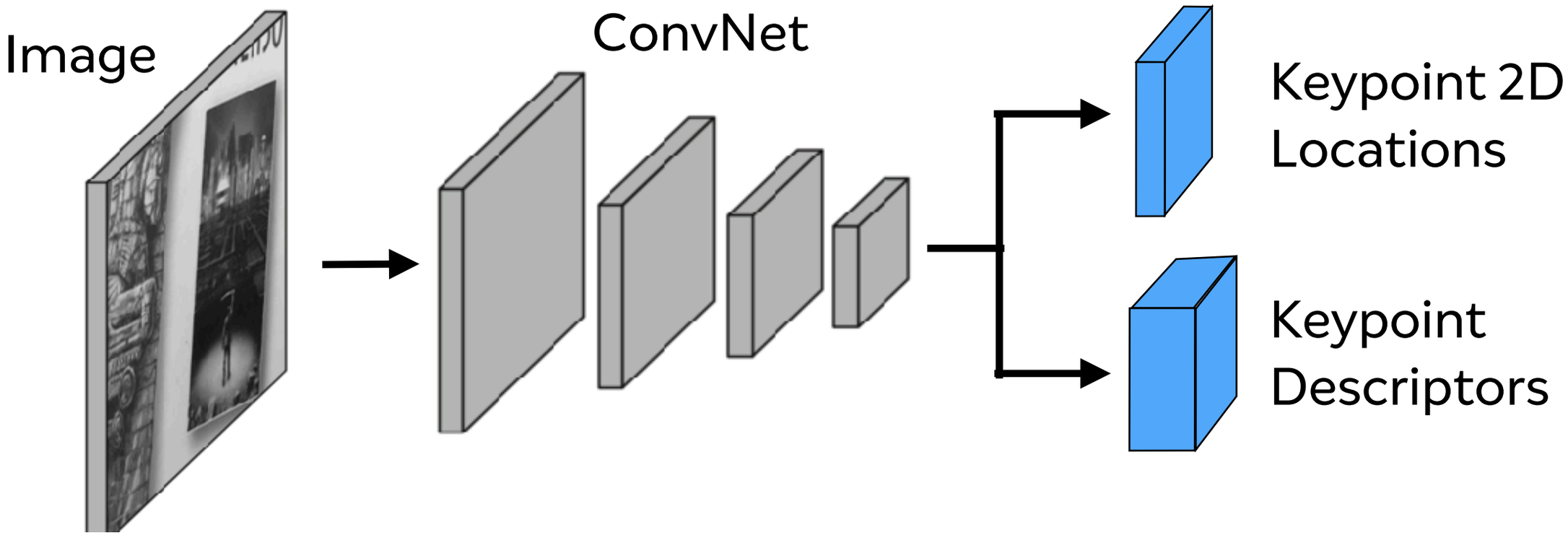



- Most low-compute Visual SLAM built on sparse frontends
- Extract points -> “Backend Ready”
- Most learned systems patch-based
  - Two separate networks
  - Lack powerful matchability of dense methods

# Question

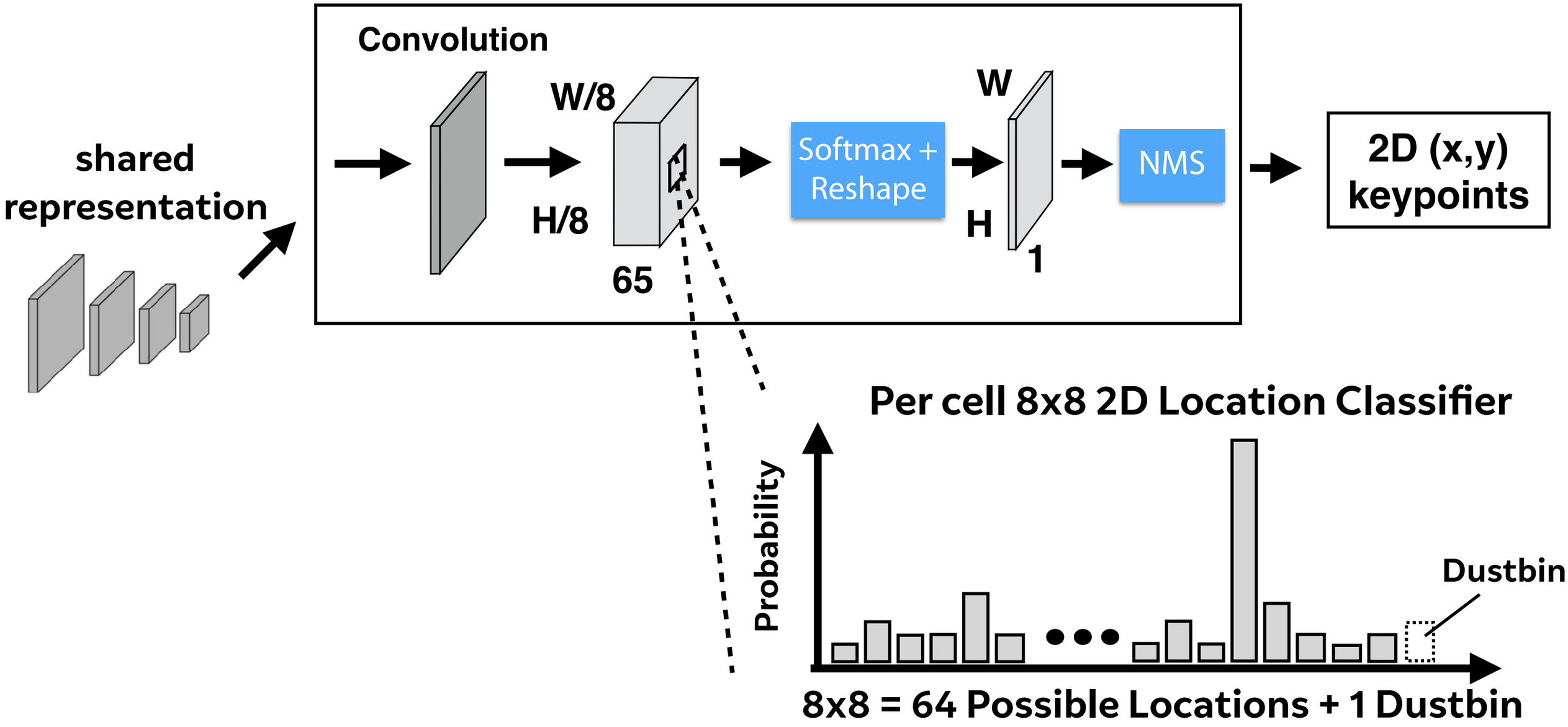
- How can we get the **power of dense matchability** and the **practicality of sparse output** in a **learnable framework**?

# SuperPoint: A Deep SLAM Front-end




- Powerful fully convolutional design
  - Points + descriptors computed jointly
  - Share VGG-like backbone
- Designed for real-time
  - Tasks share ~90% of compute
  - Two learning-free decoders: no deconvolution layers

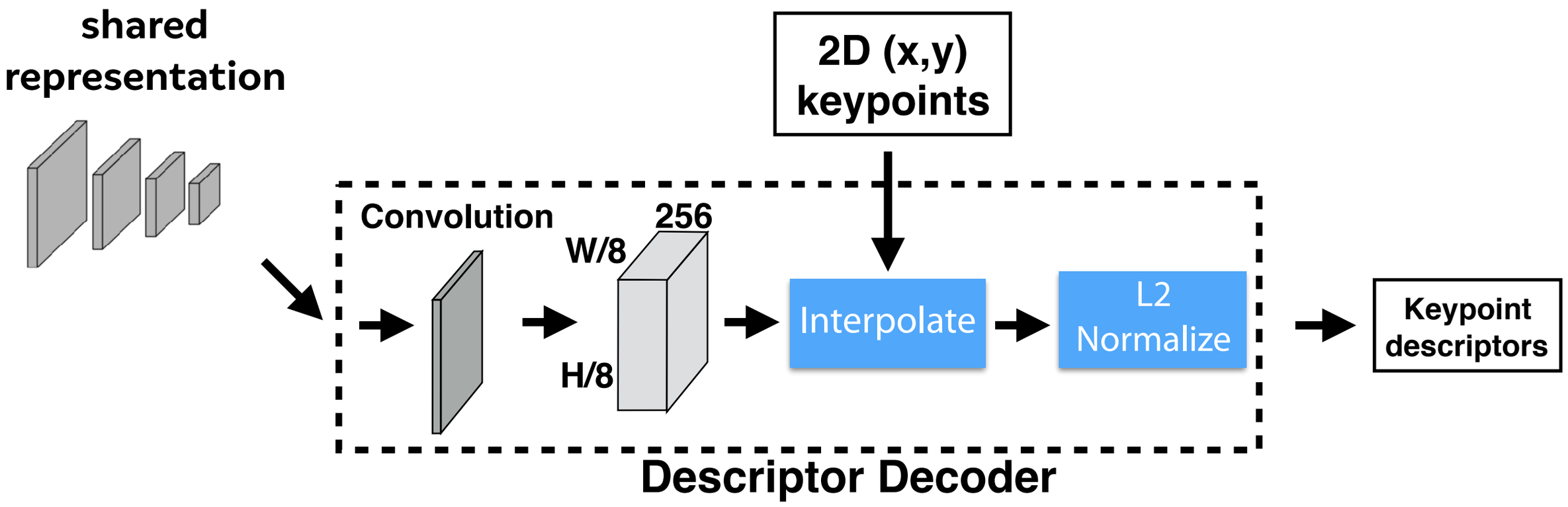
# Keypoint / Interest Point Decoder



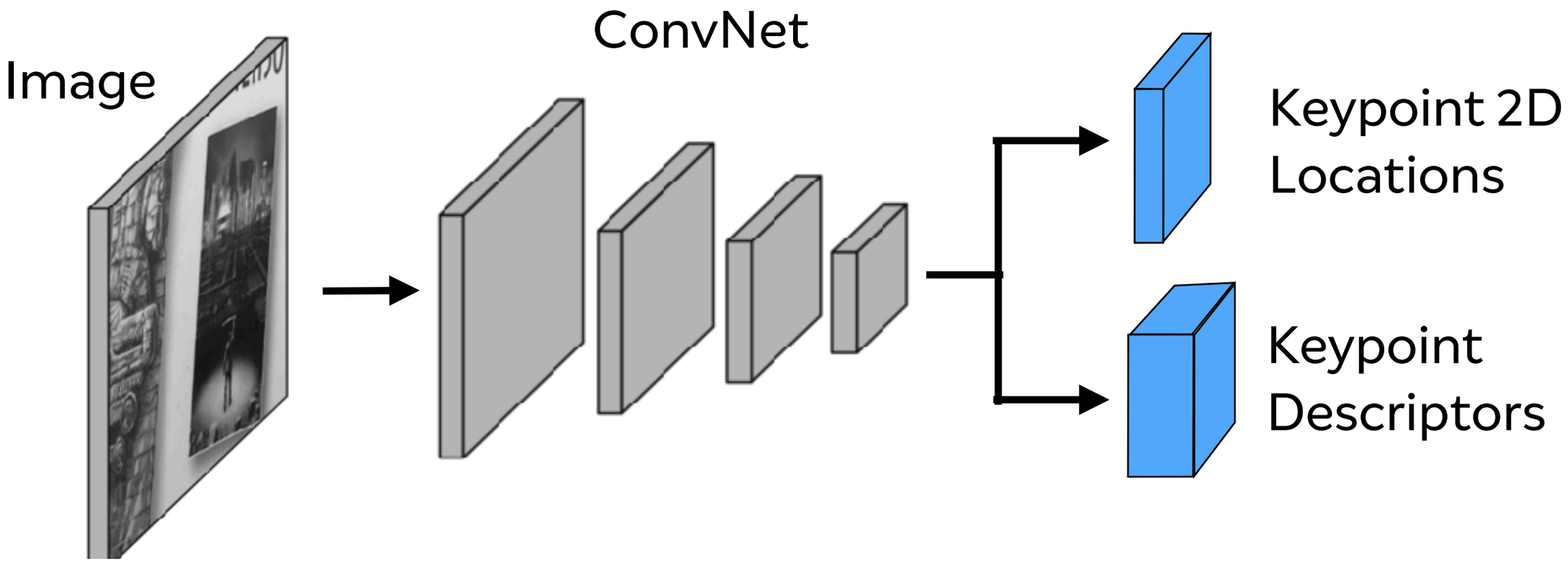

- No deconvolution layers
- Each output cell responsible for local 8x8 region

# Descriptor Decoder

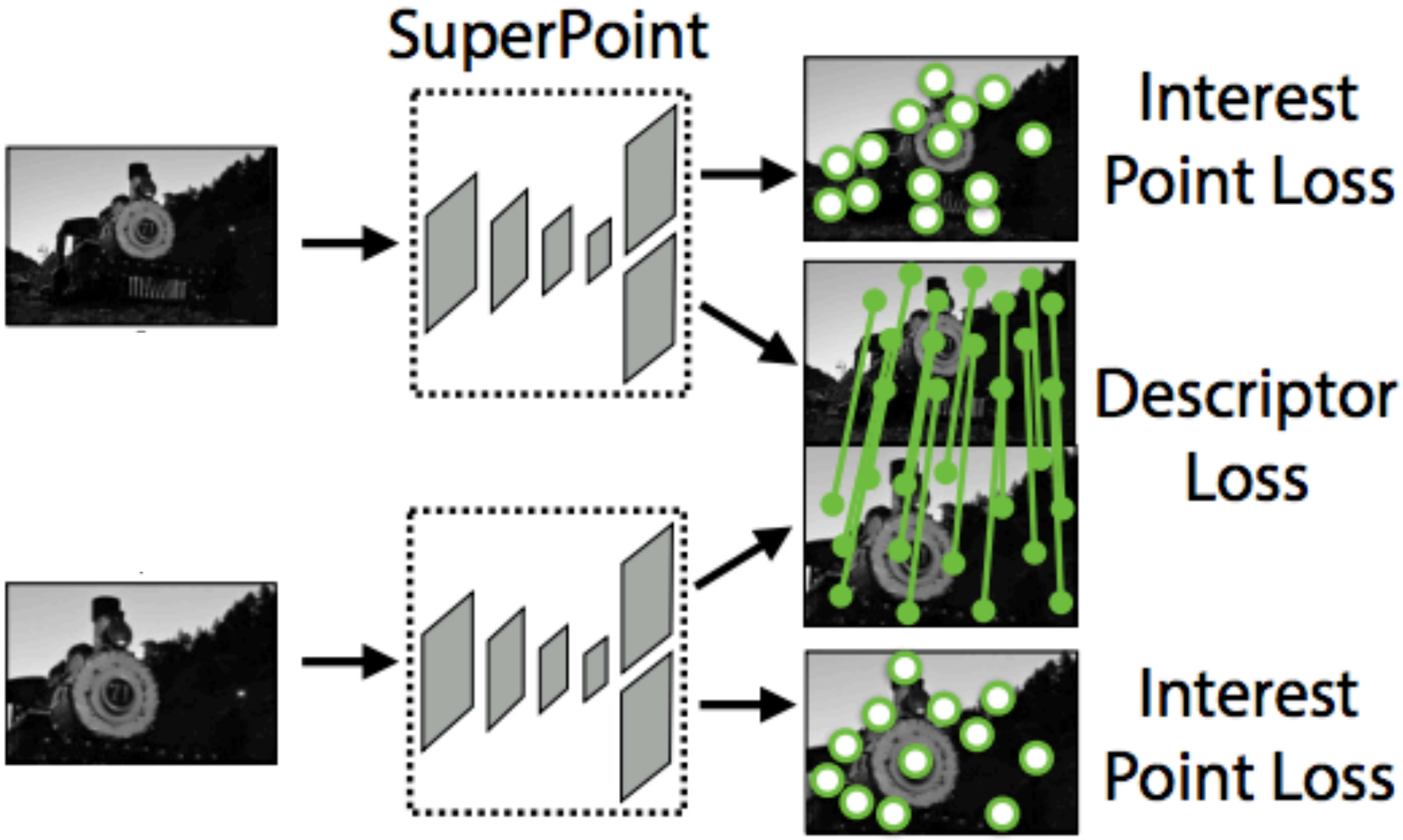
- Also no deconvolution layers
- Interpolate using 2D keypoint into coarse descriptor map



# How To Train SuperPoint?



# Setting up the Training




- Siamese training -> pairs of images
- Descriptor trained via metric learning
- Keypoints trained via supervised keypoint labels

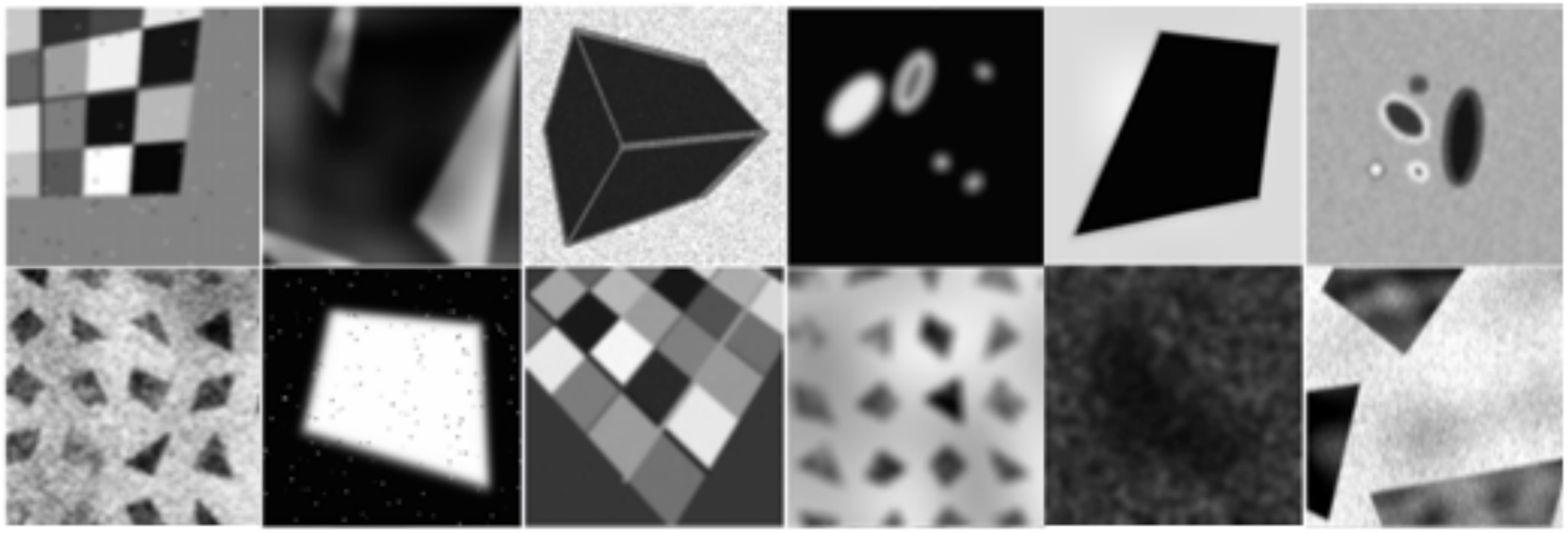
# How to get Keypoint Labels for Natural Images?



- Need large-scale dataset of annotated images
  - Too hard for humans to label

# Self-Supervised Approach

Synthetic Shapes (has interest point labels)



First train on this

"Homographic Adaptation"

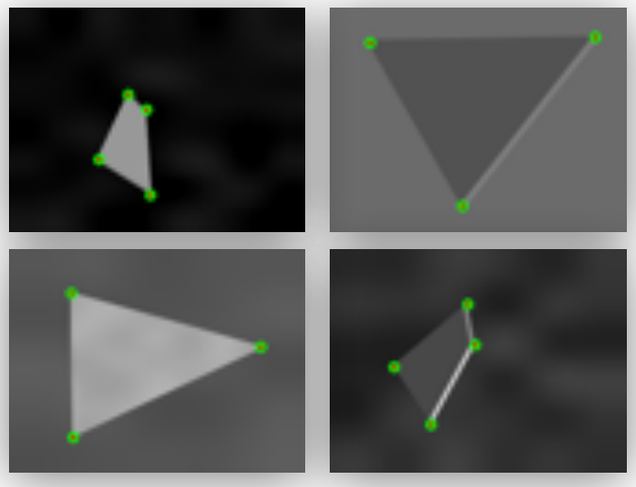
Use resulting detector to label this

MS-COCO (no interest point labels)

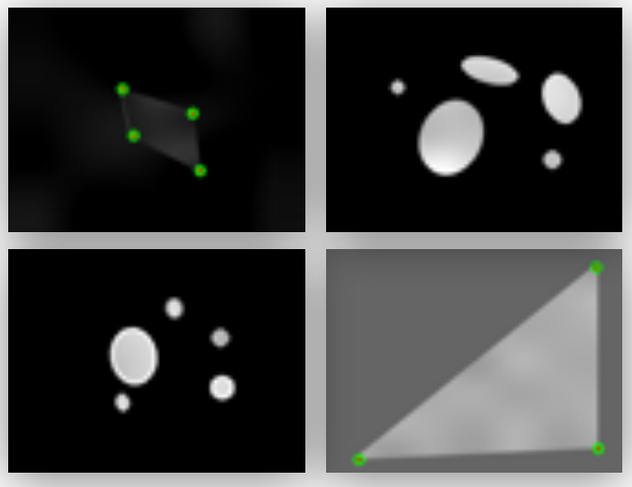


# Synthetic Training

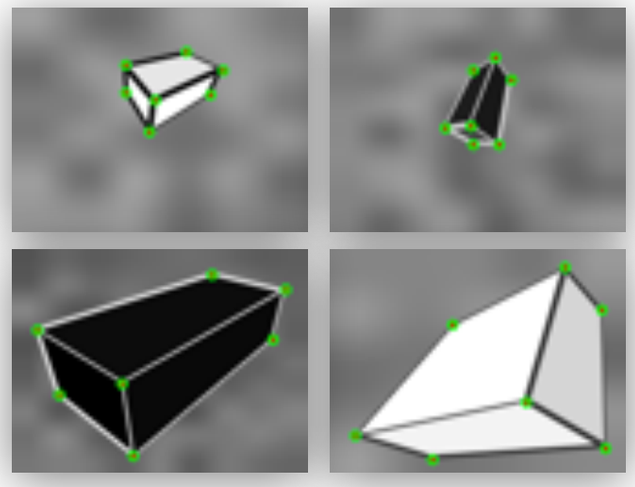
- Non-photorealistic shapes
- Heavy noise
- Effective and easy



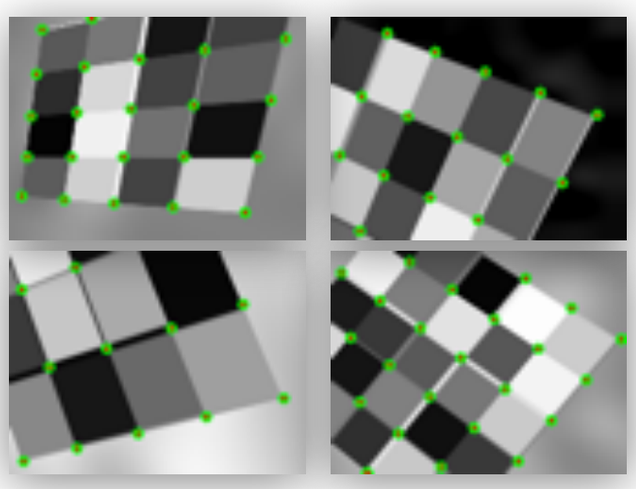
Quads/Tris



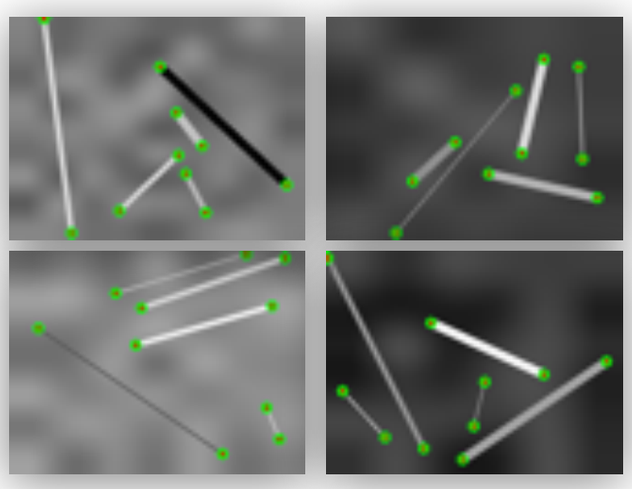
Quads/Tris/Ellipses



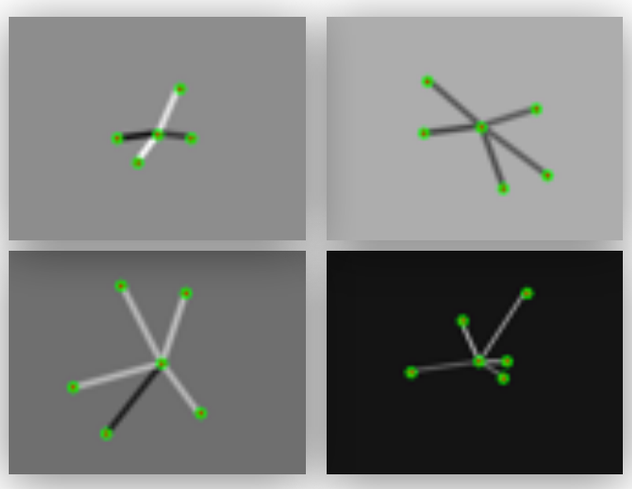
Cubes



Checkerboards



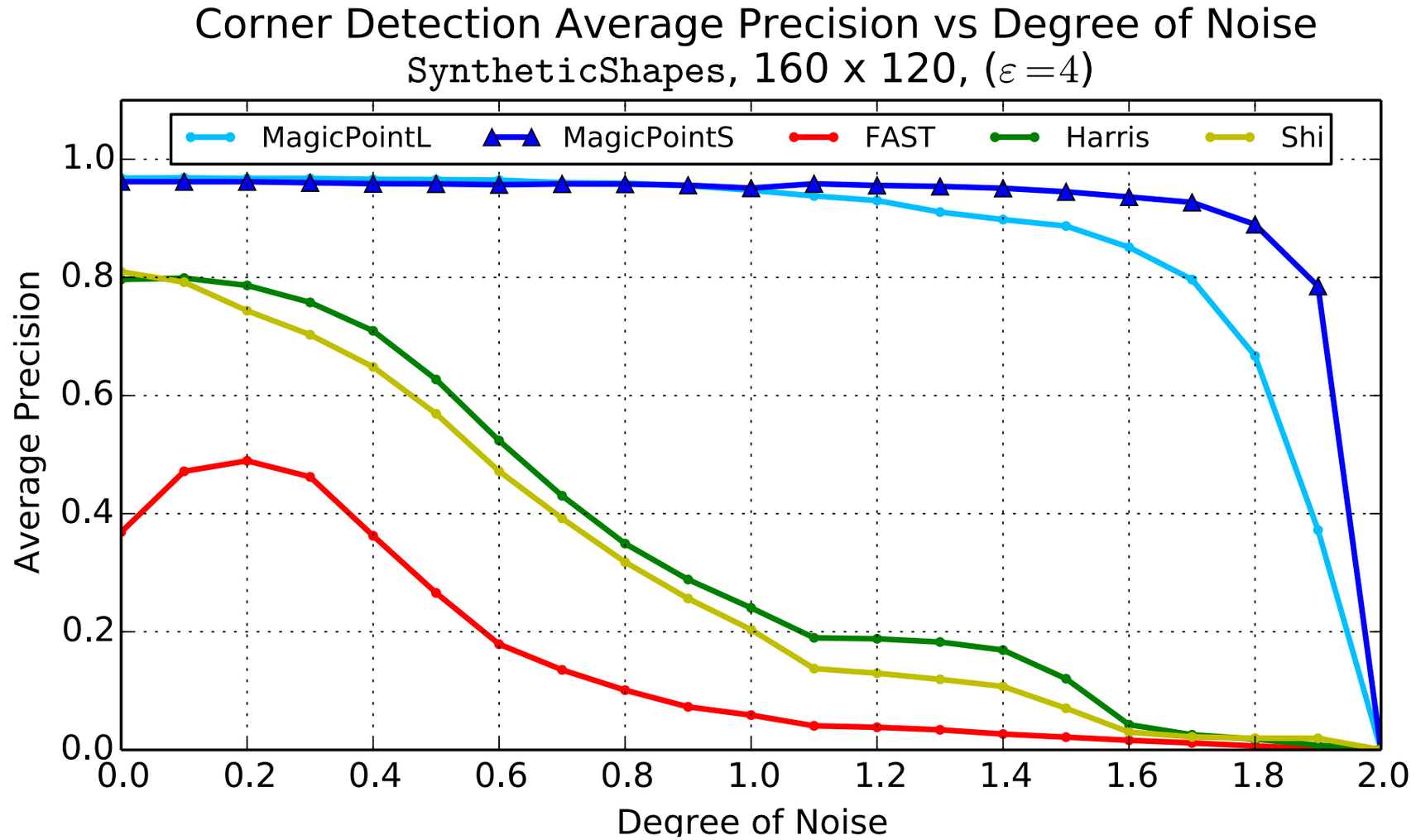
Lines



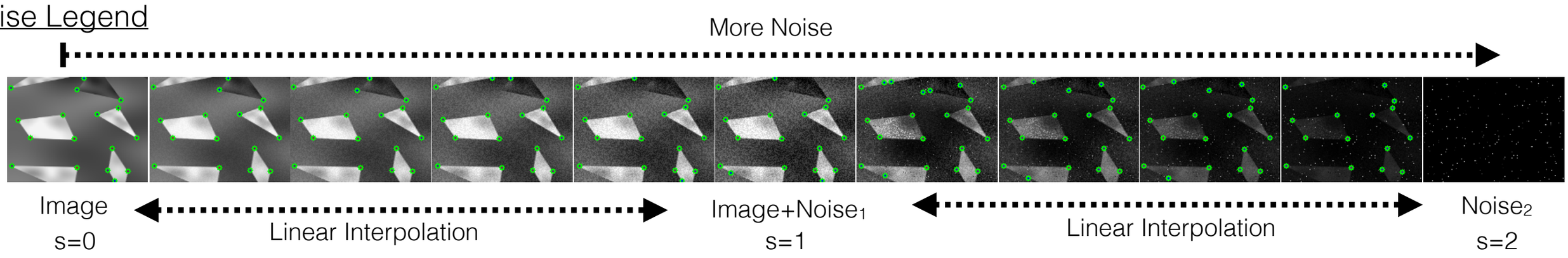
Stars

# Early Version of SuperPoint (MagicPoint)

"Toward Geometric Deep SLAM"
DeTone et. al. 2017




Noise Legend



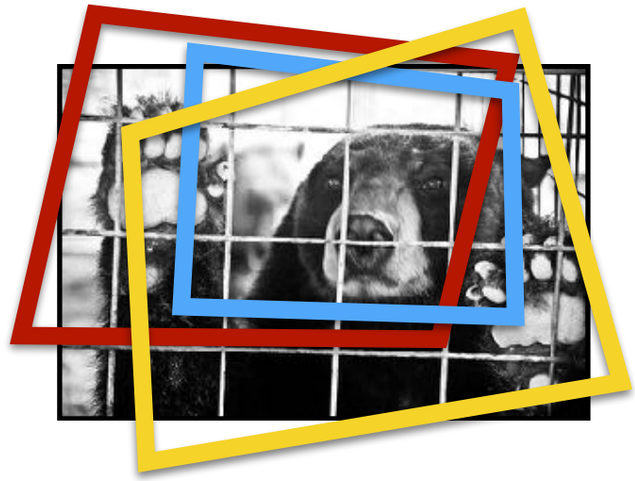
# Generalizing to Real Data

- Synthetically trained detector
  - Works! Despite large domain gap
    - Worked well on geometric structures
    - Under performed on certain textures unseen during training

**Unlabeled Input Image**



**Synthetic Warp + Run Detector**



# Homographic Adaptation



**Point Set #1**



**Point Set #2**

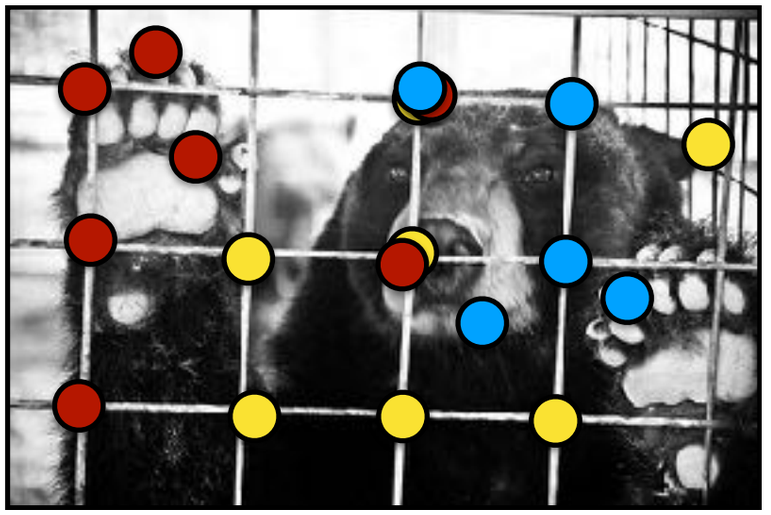


**Point Set #3**

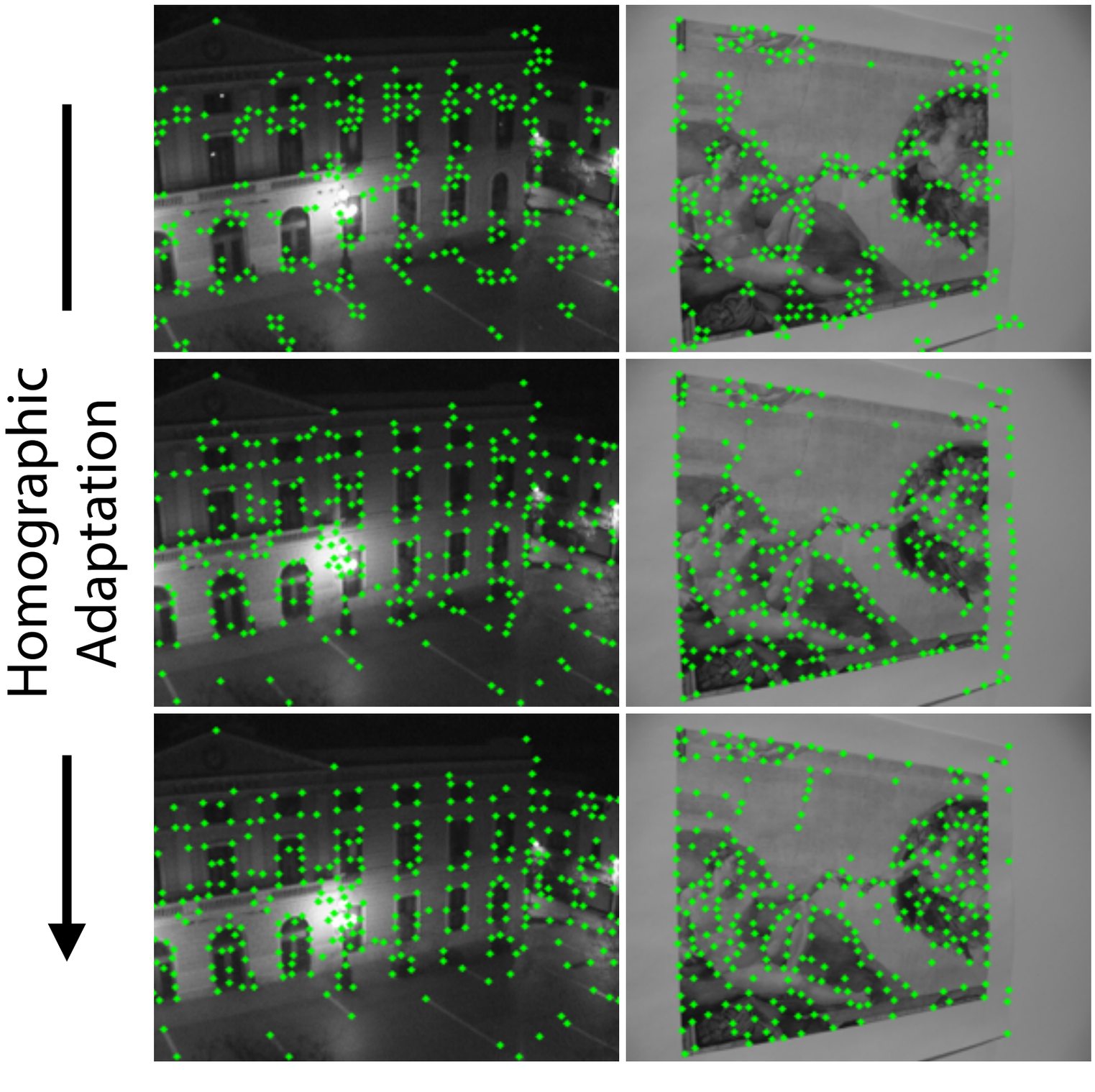
- Simulate planar camera motion with homographies
- Self-labelling technique
  - Suppress spurious detections
  - Enhance repeatable points

**Point Aggregation**

**Detected Point <u>Superset</u>**



# Iterative Homographic Adaptation




- Label, train, repeat, …
- Resulting points:
  - Higher coverage
  - More repeatable

# HPatches Evaluation

- Homography estimation task
- Dataset of 116 scenes each with 6 images = 696 images
- Indoor and outdoor planar scenes
- Compared against LIFT, SIFT and ORB

50% of dataset:
Illumination
Change





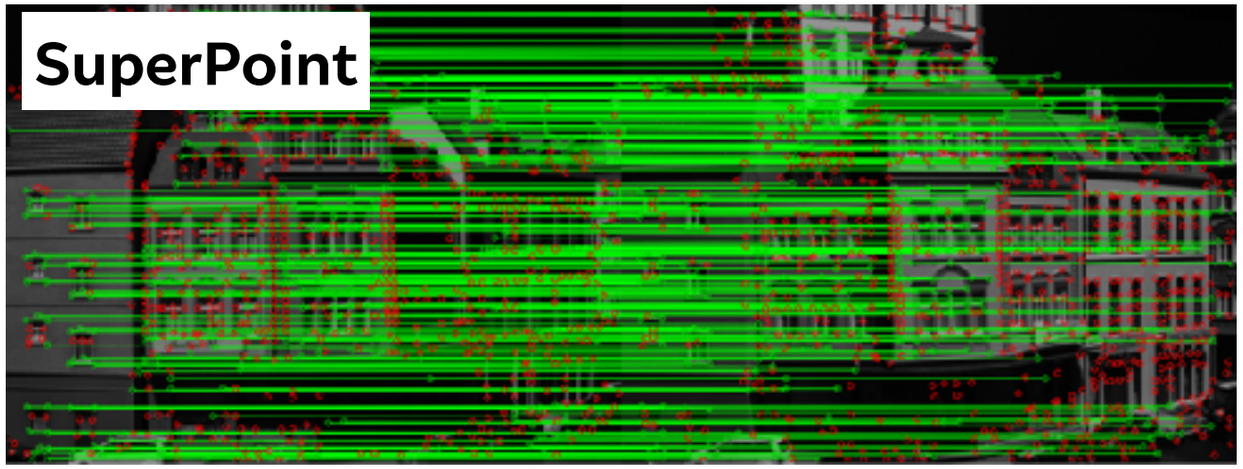
50% of dataset:
Viewpoint
Change

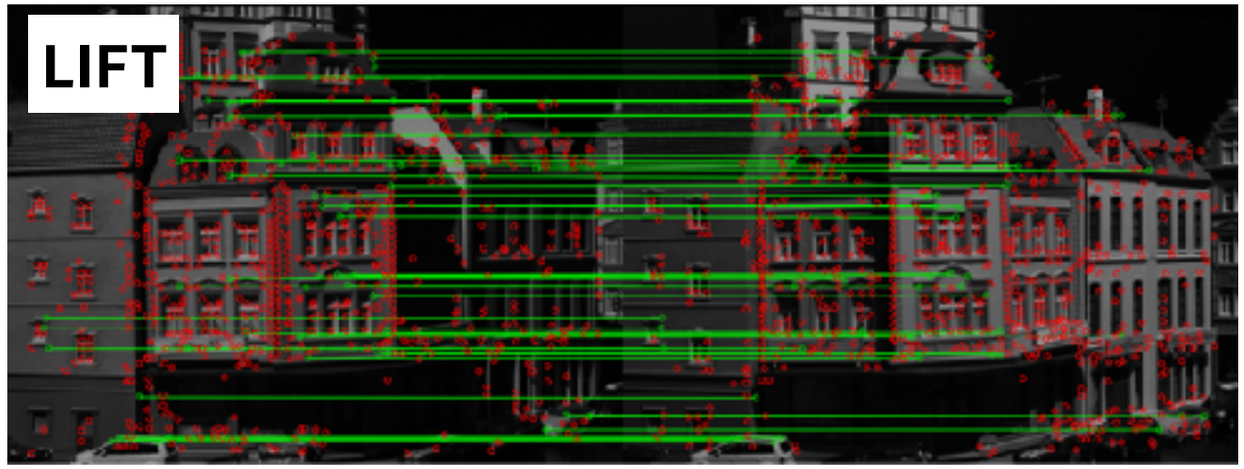


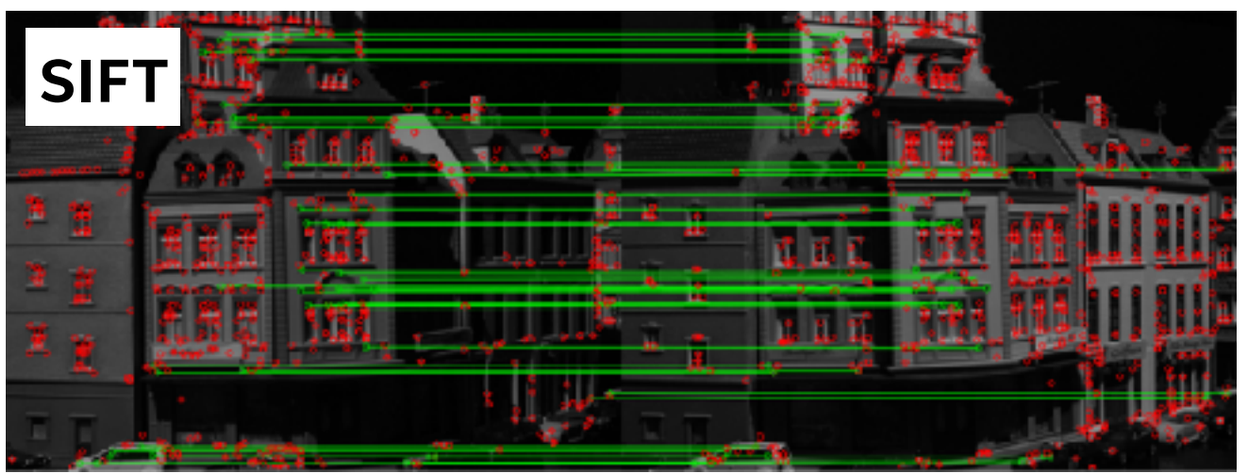


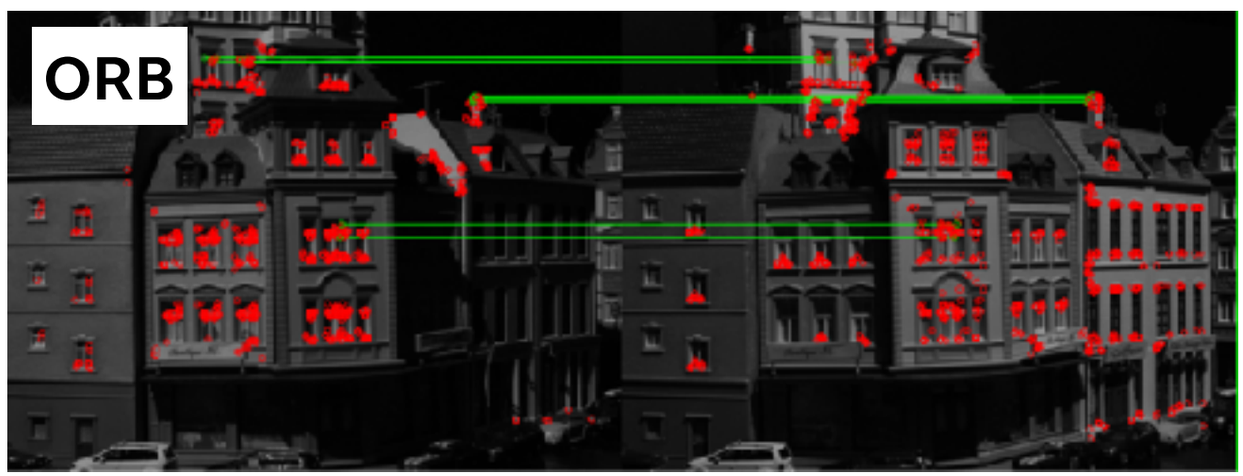
# Qualitative Illumination Example

- SuperPoint -> denser set of correct matches
- ORB -> highly clustered matches



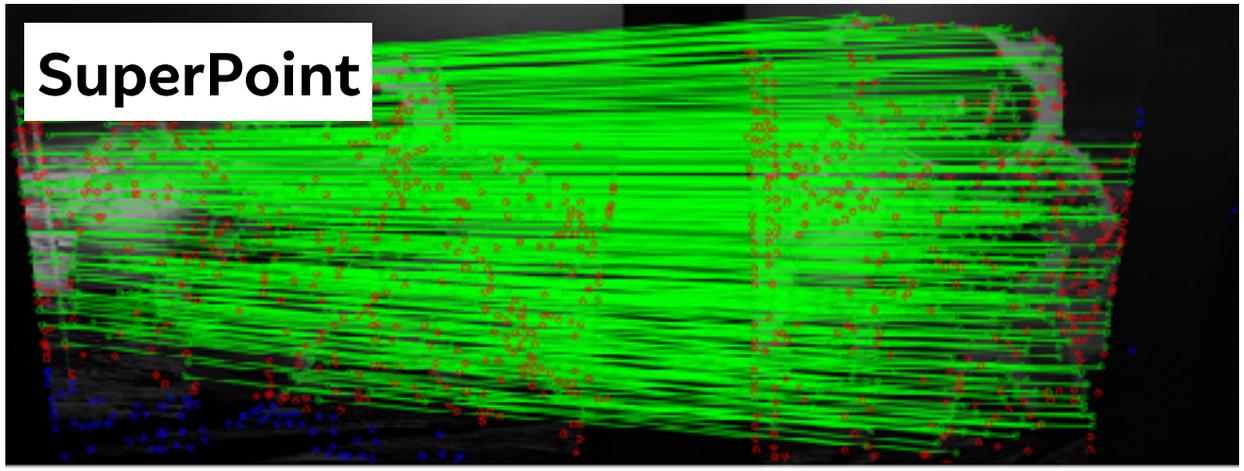



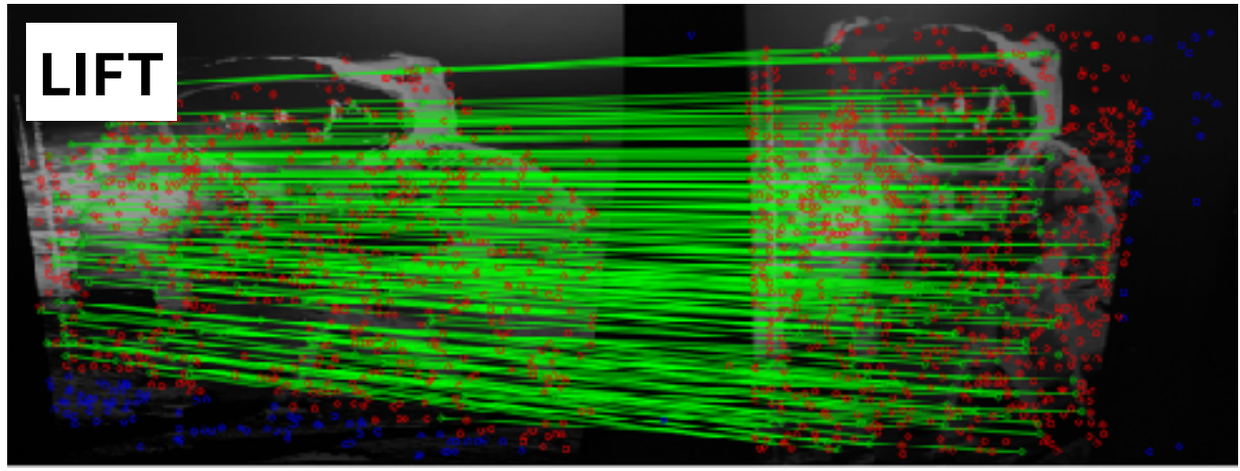



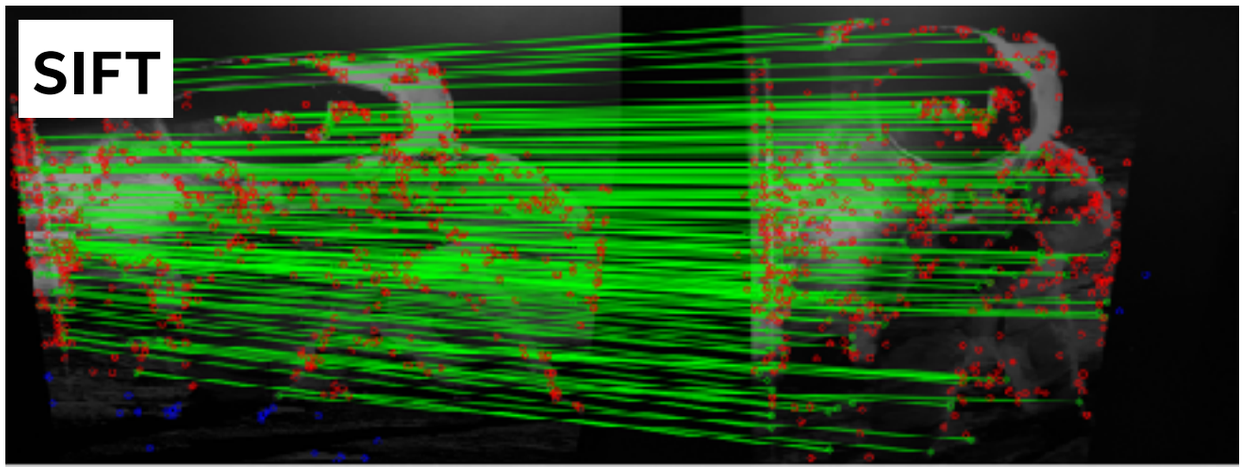



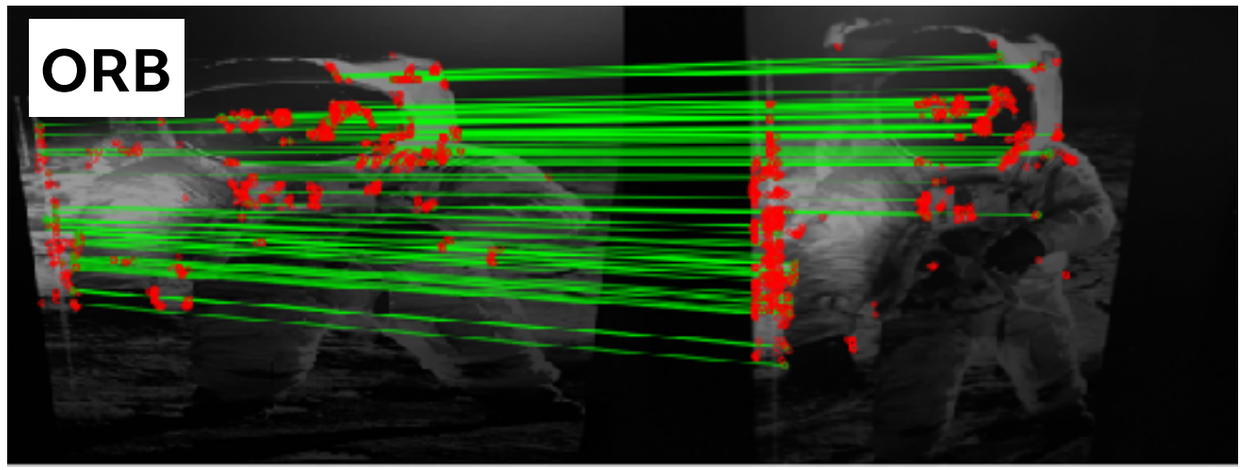
# Qualitative Viewpoint Example #1

- Similar story



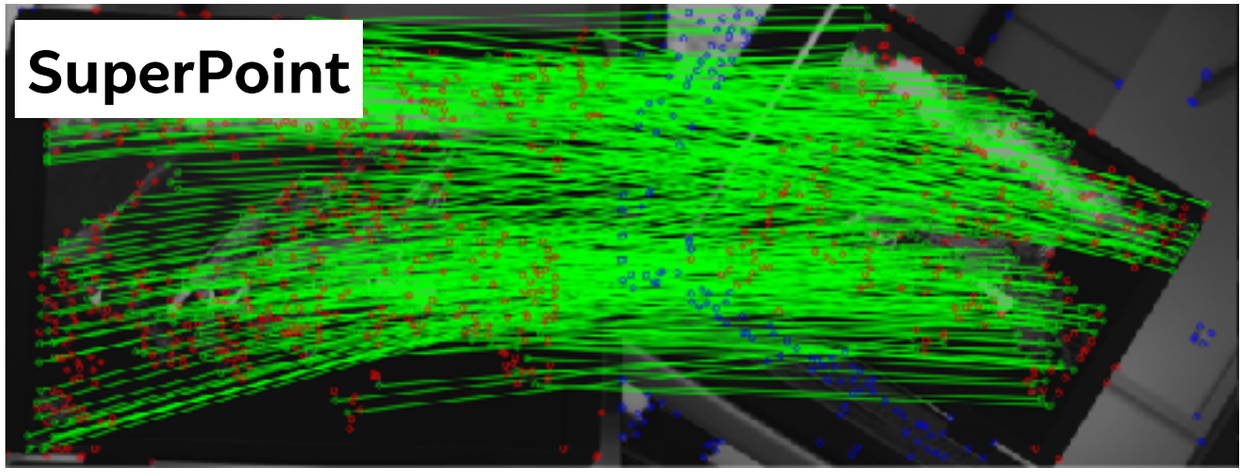



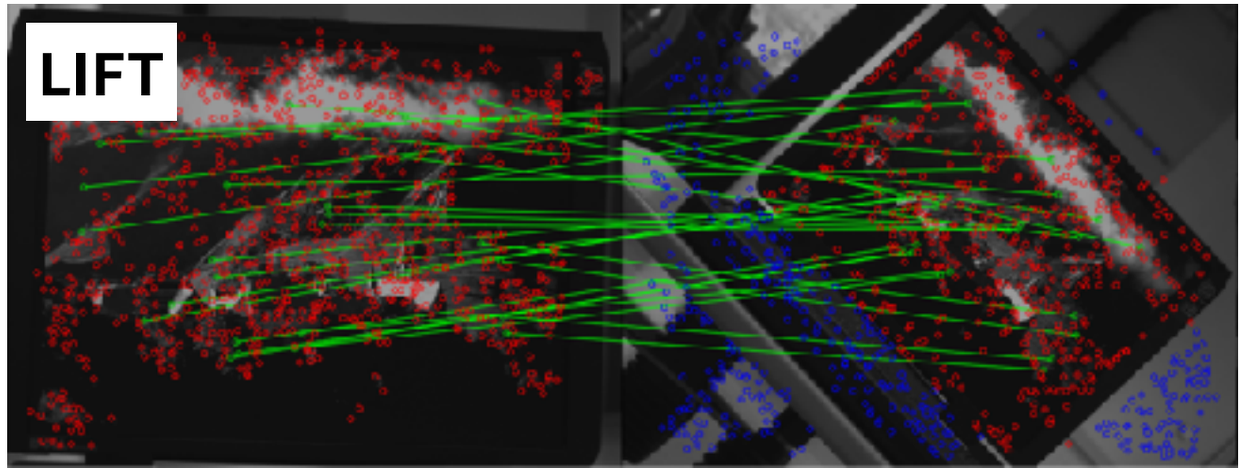



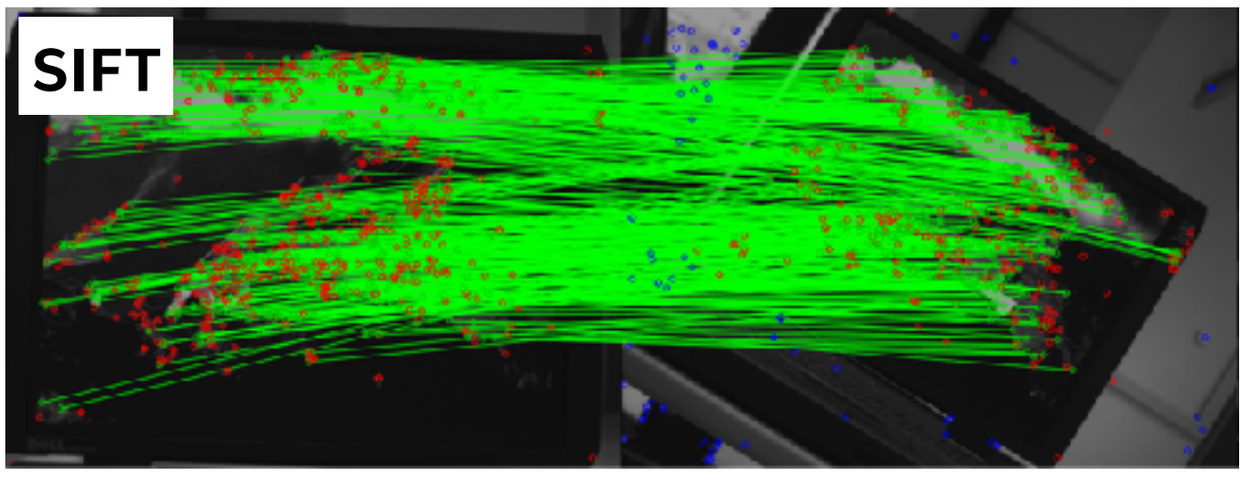



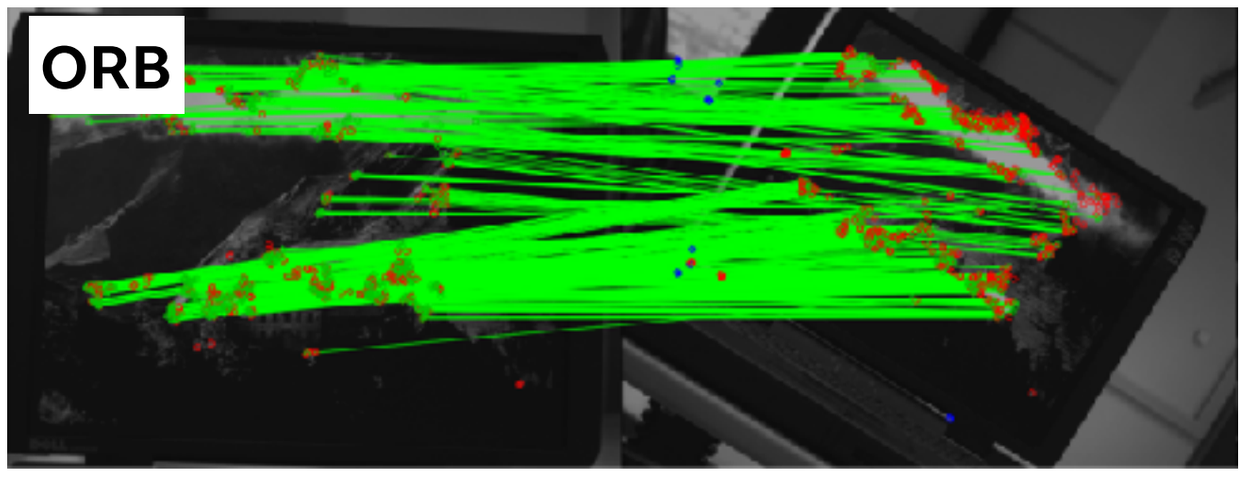
# Qualitative Viewpoint Example #2

- In-plane rotation of ~35 degrees



# HPatches Evaluation

| | Core Task | Sub-metrics | | | |
|---|---|---|---|---|---|
| | | Descriptor Metrics | | Detector Metrics | |
| | Homography Estimation ↑ | NN mAP ↑ | M. Score ↑ | Rep. ↑ | MLE ↓ |
| SuperPoint | **0.684** | **0.821** | **0.470** | 0.581 | 1.158 |
| LIFT | 0.598 | 0.664 | 0.315 | 0.449 | 1.102 |
| SIFT | 0.676 | 0.694 | 0.313 | 0.495 | **0.833** |
| ORB | 0.395 | 0.735 | 0.266 | **0.641** | 1.157 |

# Timing SuperPoint vs LIFT

- Speed important for low-compute Visual SLAM
- SuperPoint total 640x480 time: ~ 33 ms
- LIFT total 640x480 time: ~2 minutes

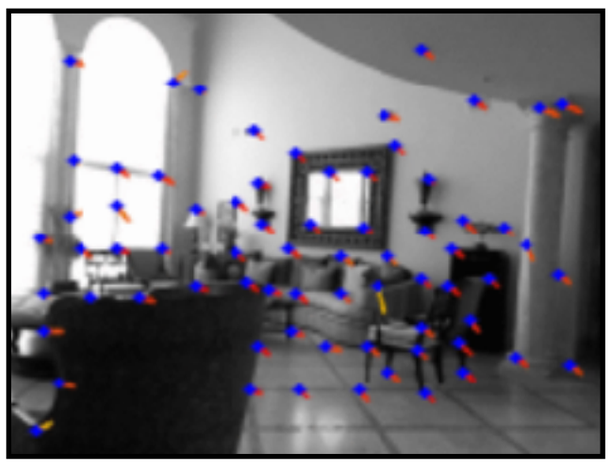
# 3D Generalizability of SuperPoint

- Trained+evaluated on planar, does it generalize to 3D?
- “Connect-the-dots” using nearest neighbor matches
- Works across many datasets / input modalities / resolutions!

*Freiburg (Kinect)*



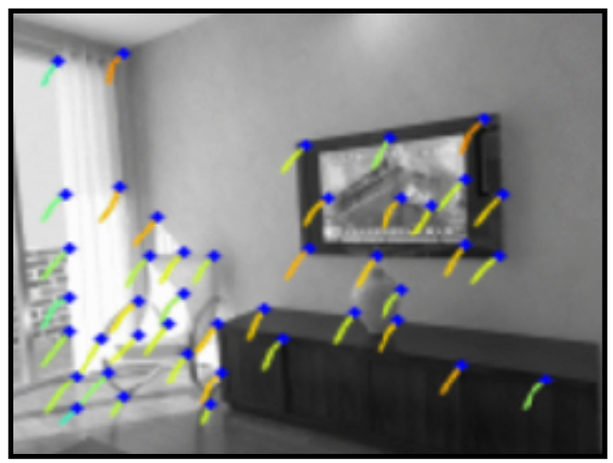
*NYU (Kinect)*



*MonoVO (fisheye)*



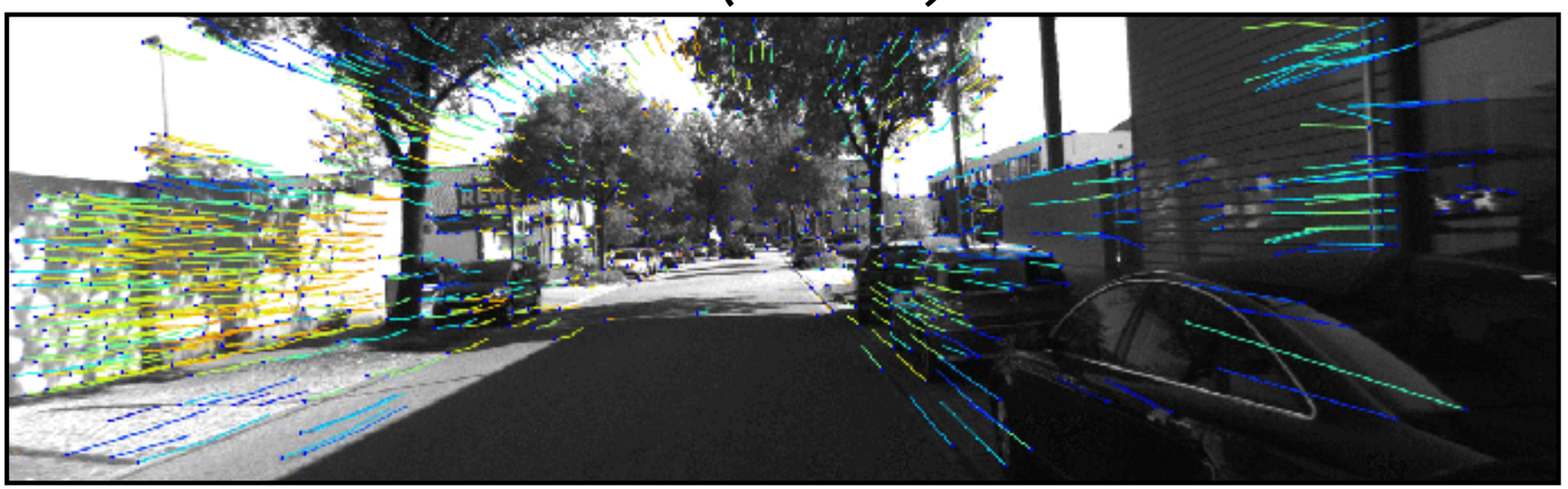
*ICL-NUIM (synth)*



*MS7 (Kinect)*



*KITTI (stereo)*



# New Announcement, Research @ MagicLeap

Public Release of Pre-trained Net:

**github.com/MagicLeapResearch/SuperPointPretrainedNetwork**

- Sparse Optical Flow Tracker Demo
- Implemented in Python + PyTorch
- Two files, minimal dependencies
- Easy to get up and running



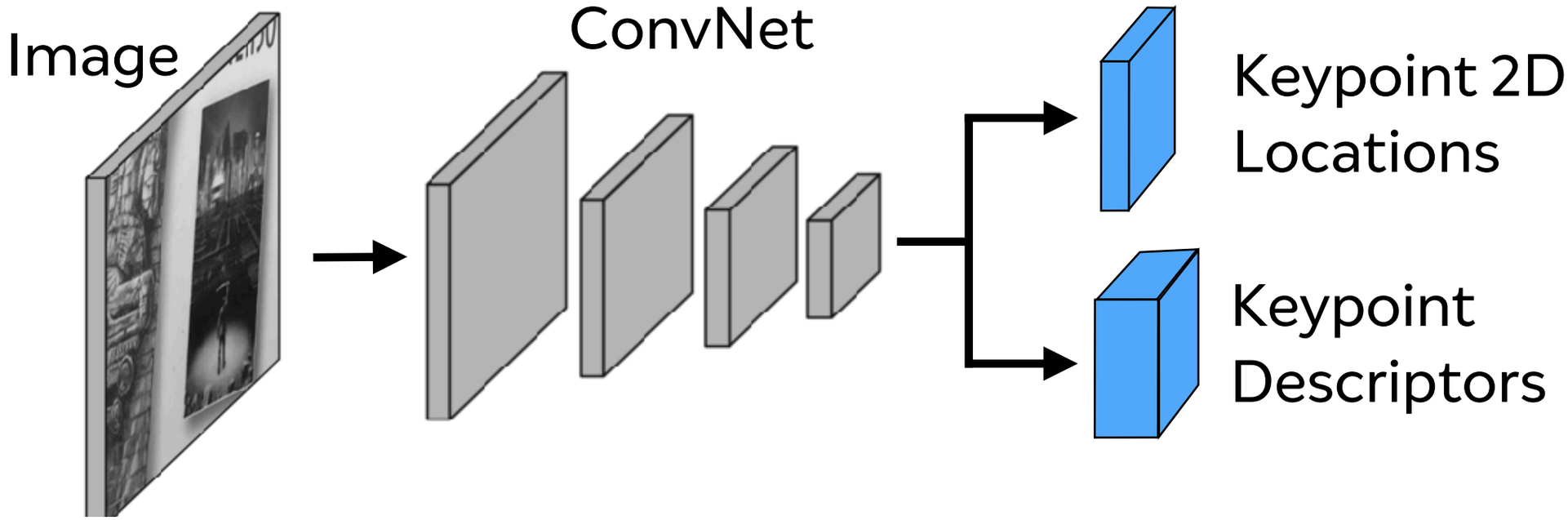
# Take-Aways

- “SuperPoint”: A Modern Deep SLAM Frontend
  - Non-patch based fully convolutional network
  - Real-time deployability
- Self-supervised recipe to train keypoints
  - Synthetic pre-training
  - Homography-inspired domain adaptation
- Public code available to run SuperPoint

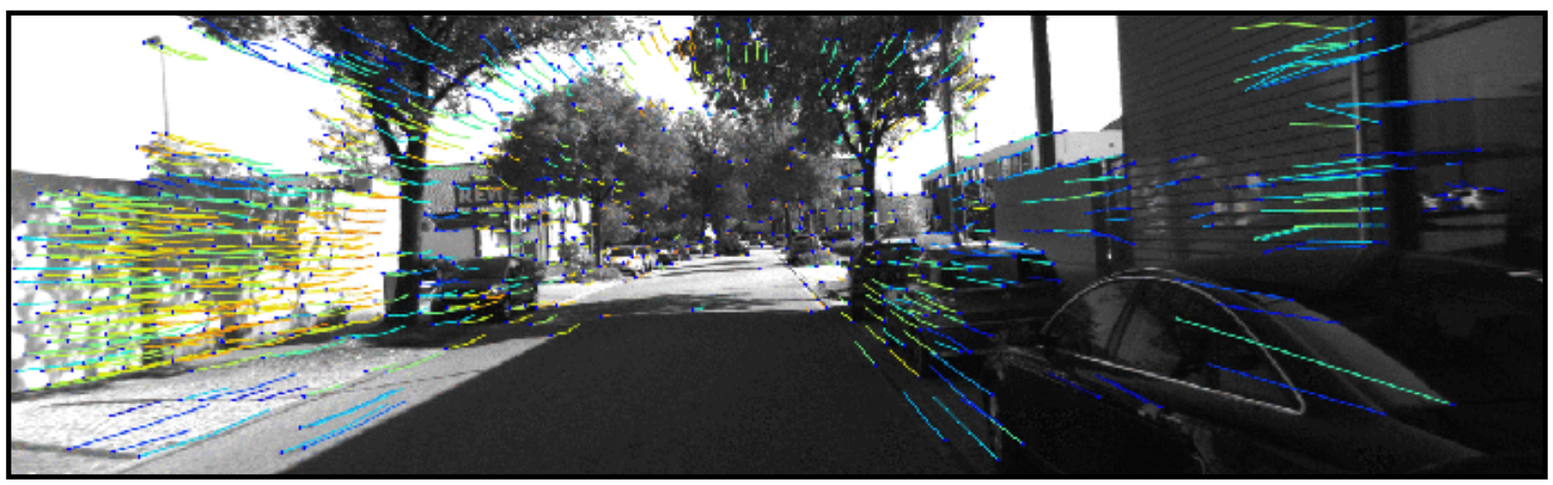
# Thank You

# Questions?

SuperPoint: A Modern Deep SLAM Front-end



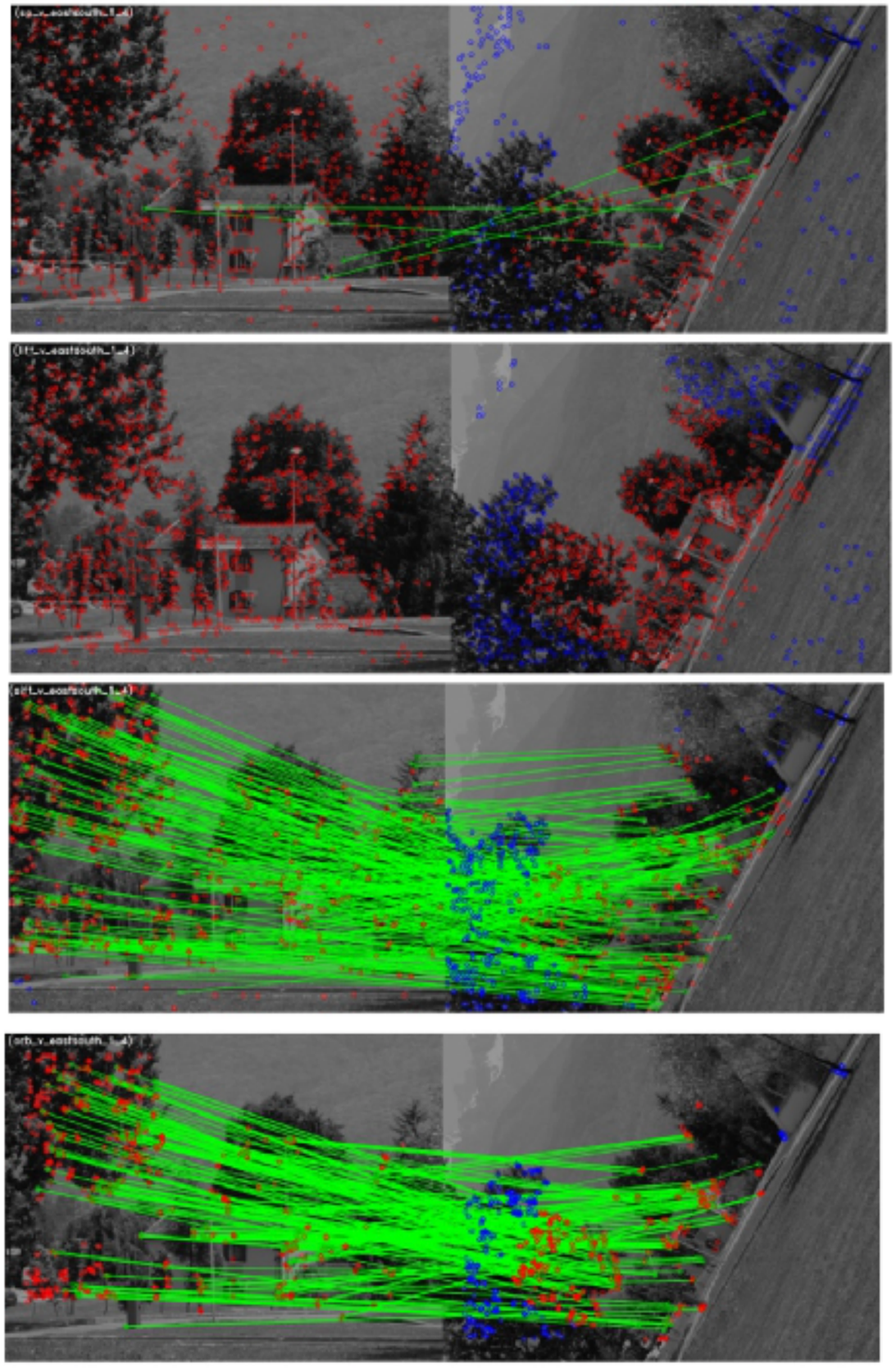


# Extra Slides

# Failure Mode: Extreme Rotation

- Extreme in-plane rotations
- Trained for ~30 deg rotations
- Optimized tracking scenarios
- LIFT also struggles, despite learned orientation estimation



**Super-Point**

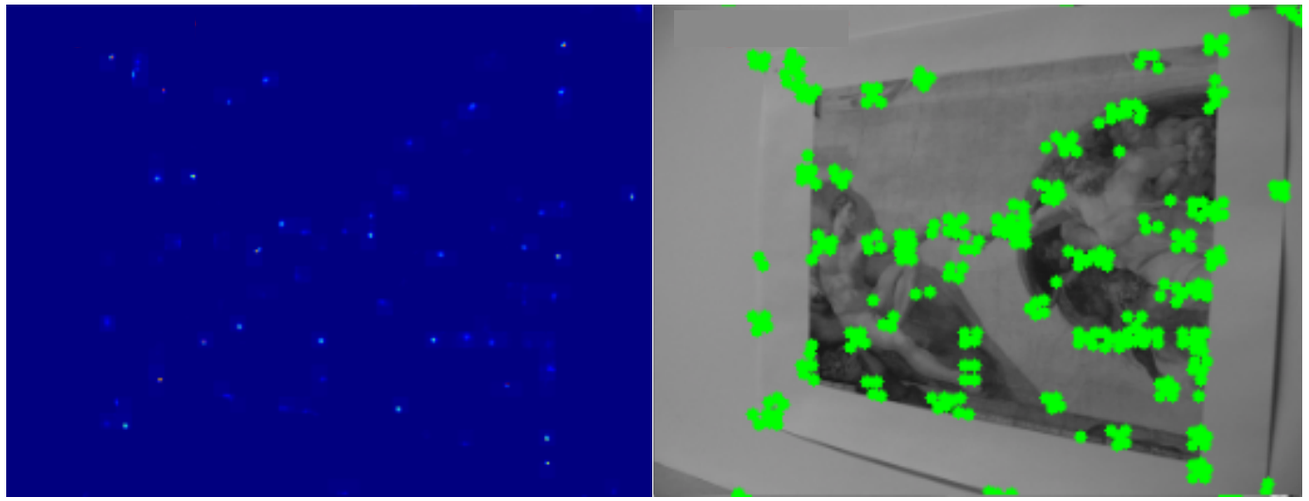
**LIFT**

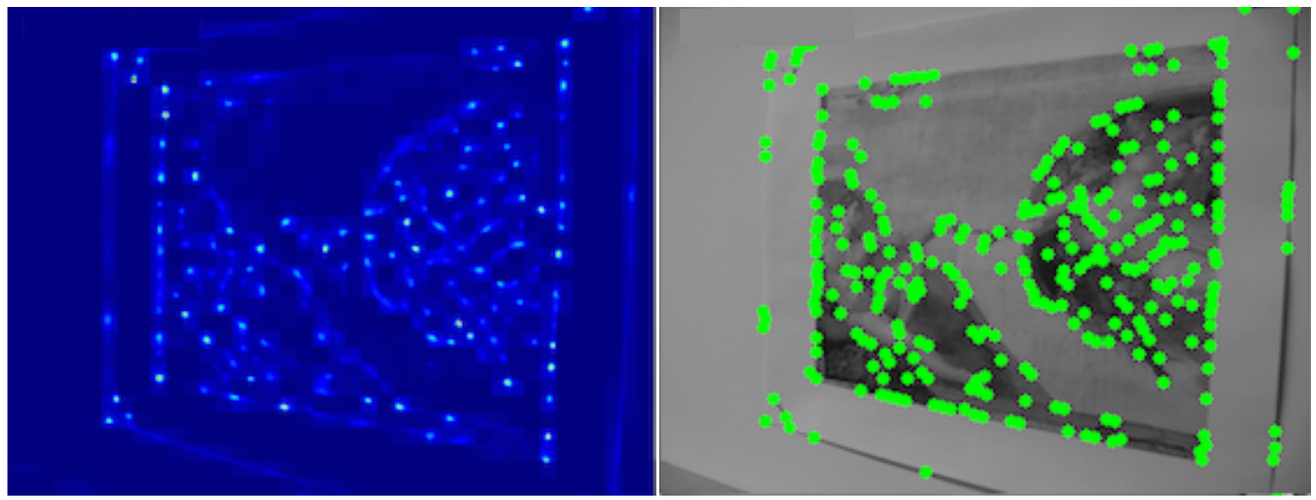
**SIFT**

**ORB**

# Iterative Homographic Adaptation

MagicPoint





SuperPoint

